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BANGLADESH UNIVERSITY OF ENGINEERING AND
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CSE 450 – CAPSTONE PROJECT

Persuasion For Good: Multi-Label Strategy Recognition and Donation-Outcome Analysis

*Multi-label persuasion-strategy recognition, donation-intent
classification,
and donation-outcome analysis on the Persuasion-For-Good corpus*

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Declaration

We hereby declare that the work presented in this report, titled **Persuasion For Good: Multi-Label Strategy Recognition and Donation-Outcome Analysis**, is the outcome of our own investigation carried out under the supervision of **[FILL IN: Supervisor’s full name and designation]**

Department of Computer Science and Engineering, BUET. Wherever contributions of others are involved, every effort has been made to acknowledge this clearly with due reference to the literature. This work has not been submitted elsewhere for the award of any degree or diploma.

The Persuasion-For-Good donation-solicitation dialogues used in this project originate from the publicly released corpus of Wang et al. (2019); our own contribution is the taxonomy, the multi-label persuasion-strategy annotation of 10,600 persuader turns, and all classification, ensembling, and statistical analysis built on top of it, as described in Chapter 4. Portions of the strategy-level and dialogue-level annotation were produced with the assistance of a large language model under a documented human-verification checkpoint (Section 4.2); this is disclosed here and again where the annotation methodology is described in full.

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We also thank the Department of Computer Science and Engineering, BUET, for providing the academic environment and computational latitude that made a project of this experimental scope possible, and Kaggle for the free GPU compute (NVIDIA T4×2 sessions) on which every model in this report was trained.

Finally, we thank the authors of the original Persuasion-For-Good corpus (Wang et al., 2019), whose public release of the raw dialogues made the entire annotation and modeling effort in this project possible.

Abstract

Persuasive dialogue is central to charitable fundraising, yet the Persuasion-For-Good corpus (Wang et al., 2019) – the largest public resource of real donation-solicitation dialogues – ships with only a binary donation outcome and a small demonstrative sample of strategy labels. It is therefore unknown, at scale, *which* persuasion strategies a persuader actually used turn by turn, and consequently unknown which strategies are associated with a donation once confounds like dialogue length are controlled for. This project closes that gap end to end: a 41-strategy / 11-category multi-label taxonomy was designed and applied to all 10,600 persuader turns of the corpus (1,017 dialogues) under an LLM-assisted, human-checkpointed annotation process; a multi-label classifier was built to recognize these strategies automatically; and the recovered strategy labels were used to model donation outcome.

The classification system evolved through seventeen measured iterations (v1–v17). Each iteration is a controlled experiment – a documented hypothesis, a change scoped to as few variables as possible, deterministic *sentinel* ensemble members whose scores must reproduce bit-for-bit between an iteration and its predecessor before any comparison is trusted, and a decision recorded whether the change survived contact with held-out data. Several plausible ideas were rejected this way (target-span attention pooling, prior-initialized classification-head biases, a naive class-imbalance loss reweighting that improved micro- but not macro-F1); two were kept (an architecture-dependent context-width effect, and an empirical-Bayes threshold-shrinkage decision rule that this project contributes). The most consequential single change was reframing the task itself: rather than predicting all 41 fine-grained strategies and reading off the 11 coarse categories as a byproduct, the final system (v15–v17) makes the 11-category task the primary training and decoding objective, with the 41-strategy task retained as an auxiliary head whose gradient demonstrably regularizes the rare categories. On the held-out test split, this raises category-level micro-F1 to **0.706** and macro-F1 to **0.564** – within 0.5 pp of the corpus’s own cross-annotator agreement ceiling on the same task (κ -analog consensus bound 0.793), meaning the remaining error is close to the noise floor of the human labels themselves, concentrated almost entirely in three categories that make up 3.4% of all positive labels.

Feeding these recovered strategy labels into a McFadden pseudo- R^2 logistic-regression outcome model raises cross-validated explained variance from 0.12 (categories alone) to **0.165** (with strategy co-occurrence features) on the same

1,017 dialogues – roughly double the 0.015–0.08 range reported by prior single-label work on a related corpus, though this project is explicit that the comparison is confounded by more than the labeling scheme and should not be over-read. A companion donation-*intent* classifier built by the same team, using a dual-stream cross-attention architecture over persuader/persuadee turns, separately reaches **89.5%** accuracy and **0.862** macro-F1 on binary donation intent, and the same architecture transfers to a structurally similar buyer–seller negotiation domain at **97.3%** accuracy.

Beyond the numbers, this report documents a reproducible experimental methodology for iterating on a multi-label classifier under a hard compute budget (a single free Kaggle GPU session per run): GPU-free smoke tests that exercise every code path on a data subsample before committing to a multi-hour run, redundant training across two Kaggle accounts to survive weekly GPU-quota exhaustion, and a statistically honest reporting convention – cluster bootstrap confidence intervals over dialogues rather than turns, dual micro-/macro-optimal decision thresholds fitted by grouped cross-validation, and an explicit annotation-reliability ceiling reported alongside every headline score so that a result is never presented as better, or worse, than the data can actually support.

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Chapter 1

Introduction and Problem Statement

1.1 Motivation

Persuasion is the mechanism by which a charity turns a stranger’s attention into a donation. Every fundraising call, every solicitation script, and every automated donation-request chatbot is, underneath, a sequence of persuasive moves – appeals to empathy, statements of credibility, offers of reciprocity, statements of urgency – chosen (consciously or not) to move a specific person toward a specific decision. Two questions follow immediately from that observation, and both matter well beyond charity:

1. **Recognition.** Given the text of a persuasive dialogue, can a system automatically identify *which* persuasion strategies are being used, turn by turn, at a level of granularity fine enough to be useful?
2. **Consequence.** Once strategies can be recognized at scale, do some of them actually predict a better outcome than others – and can that be established with enough rigor to be trusted, rather than merely correlated with confounds like dialogue length or persuader skill?

Answering these questions well has direct value for nonprofit fundraising (coaching material, script optimization, live-call feedback), for conversational-AI systems that must persuade responsibly (donation bots, negotiation agents, health-behavior nudging), and for computational social science, which has long theorized about persuasion (Cialdini, 1984) but rarely had a large, turn-level, multi-label-annotated corpus to test those theories against.

1.2 The Persuasion-For-Good Corpus and Its Limitation

The *Persuasion-For-Good* (P4G) corpus (Wang et al., 2019) is the largest public resource of real donation-solicitation dialogues: crowd-workers were paired as a *persuader*, who is asked to convince a *persuadee* to donate part of a small task

payment to the charity *Save the Children*, and a *persuadee*, who is free to donate any amount from \$0 up to the full payment. The corpus records the full dialogue transcript and the final binary/graded donation outcome for over a thousand conversations.

What the public release does *not* provide is fine-grained, exhaustive, turn-level labeling of *which* persuasion strategy each persuader turn uses. The original paper annotates only a small demonstrative subset with a coarse strategy label. This is the gap the project addresses directly: without knowing which strategies actually occur, and how densely they co-occur within a single turn, it is not possible to build either an accurate automatic strategy recognizer or a trustworthy strategy-to-outcome model.

1.3 Problem Statement

This project is organized around four concrete sub-problems, each addressed by a distinct module of the system described in Chapter 4:

P1 – Strategy taxonomy and annotation. No public, exhaustive, multi-label persuasion-strategy annotation of the P4G corpus exists. A taxonomy has to be designed, and every one of the 10,600 persuader turns across 1,017 dialogues has to be labeled with the *set* of strategies it uses (a single turn routinely uses two or three strategies at once – treating this as single-label classification would be a modeling error, not a simplification).

P2 – Automatic multi-label strategy recognition. Given a persuader turn (and its recent dialogue context), automatically predict which of the 41 fine-grained strategies (grouped into 11 higher-level categories) are present, under severe class imbalance – some strategies occur only a handful of times in the entire corpus – and under an annotation process that is itself imperfect and must be measured, not assumed perfect.

P3 – Donation-outcome analysis. Given the strategy labels for a dialogue (gold, to avoid contaminating the outcome model with the classifier’s own errors), estimate which strategies and strategy combinations are statistically associated with donation, and by how much, in a way that is honest about the limits of an observational, non-randomized corpus of this size (1,017 dialogues).

P4 – Donation-*intent* classification and generalization. As a complementary and more directly deployable capability, classify whether a *dialogue as a whole* ends in a donation intent (and of what type – outright, deferred, or conditional), and test whether an architecture built for this closely related task transfers to a structurally similar but distinct persuasive domain (buyer–seller price negotiation).

1.4 Scope and Objectives

The concrete, measurable objectives set for this project were:

- Design an 11-category / 41-strategy multi-label persuasion taxonomy covering the range of tactics observed in the P4G corpus, and apply it to every persuader turn in the first 10,600 turns of the corpus (Section 4.2).
- Build a multi-label classifier for this taxonomy that is evaluated honestly against the corpus’s own annotation-reliability ceiling, not against a naive upper bound of 1.0 (Section 5.2).
- Push the classifier’s macro-F1 – the metric that is dragged down by the rarest strategies – as high as is actually achievable given that ceiling, using a disciplined, hypothesis-driven, reproducible experimental process rather than undirected hyperparameter search (Chapter 5).
- Use the resulting (gold) strategy labels to quantify, with corrected standard errors and multiple-testing control, how much of the variance in donation outcome the strategy composition of a dialogue explains, and compare a multi-label specification against a single-label projection of the same data to isolate whether multi-labeling itself adds predictive value (Section 5.6).
- Build and evaluate a companion donation-intent classifier, and test whether its architecture generalizes to a different persuasive domain (Section 5.8, Section 5.9).

The remainder of this report is organized as follows. Chapter 2 situates the project against prior work on persuasion taxonomies, the P4G corpus, multi-label classification under class imbalance, and outcome modeling from text. Chapter 3 states the functional and non-functional requirements the system was designed against. Chapter 4 describes the full system architecture, including the annotation methodology and the classifier’s seventeen-iteration design history. Chapter 5 reports and interprets every result, with confidence intervals and significance tests throughout. Chapter 6 covers project management, tooling, and ethical considerations, and Chapter 7 summarizes contributions, limitations, and future work.

Chapter 2

Background and Related Work

2.1 Persuasion Theory and Strategy Taxonomies

The social-psychology literature on persuasion long predates NLP work on the topic. Cialdini’s six principles of influence – reciprocity, commitment and consistency, social proof, authority, liking, and scarcity (Cialdini, 1984) – are the most widely cited organizing framework, and their fingerprints are visible in the taxonomy this project designed (Section 4.2.1): *Reciprocity and Exchange*, *Commitment and Consistency*, and *Urgency and Scarcity* are category names taken almost directly from Cialdini, while *Social Influence* and *Credibility Appeal* correspond to social proof and authority respectively. The taxonomy extends this classical framework with categories specific to charitable solicitation dialogue that do not map cleanly onto Cialdini’s six – *Rational Appeal*, *Emotional Appeal*, *Framing and Presentation*, *Threat and Pressure*, and *Information Provision* and *Relational and Interactive* moves that are largely conversational scaffolding rather than persuasive content per se.

2.2 The Persuasion-For-Good Corpus

Wang et al. (2019) introduced the P4G corpus and, alongside it, an 11-strategy annotation scheme applied to a small labeled subset, together with an early sequence-model strategy classifier and a persuader-side dialogue system trained to imitate high-donation-rate human persuaders. Their release established the corpus this project builds on, but three limitations motivated the present work: the strategy annotation covers only a fraction of the corpus; each turn was treated as carrying at most one dominant strategy, whereas manual inspection of the corpus shows a turn routinely combines two or three simultaneously (Section 4.2.4); and no systematic, statistically corrected analysis was published of which strategies are actually associated with the final donation decision once dialogue-level confounds are accounted for.

Subsequent work using P4G has largely focused on the intent-prediction side of the corpus (predicting whether a dialogue will end in a donation) rather than the

fine-grained strategy side, which is the gap this project’s Module B (Chapter 4) is built to fill.

2.3 Multi-Label Text Classification Under Class Imbalance

Framing strategy recognition as multi-label rather than single-label classification is not a stylistic choice; the empirical labels—a mean of 1.68 strategies and 1.50 categories per turn, with under 7% of turns carrying zero labels (Section 4.2.4)—rule out a single-label formulation on this corpus. Binary cross-entropy per label is the standard baseline but treats every label symmetrically, which is a poor fit when, as here, the label distribution spans nearly three orders of magnitude (the rarest strategy occurs twice in the corpus, the most common occurs 2,860 times). Ben-Baruch et al. (2021) propose Asymmetric Loss (ASL), which decouples the focusing power applied to positive and negative predictions and adds a small negative-probability margin so that easy negatives contribute almost nothing to the gradient; this project adopts ASL as its base loss (Section 4.4.2) and additionally makes the per-label positive weighting configurable per ensemble member, since it was measured to trade micro-F1 against macro-F1 rather than improving both (Table 5.11).

Resampling is the more traditional response to label imbalance, and is what this project’s own EDA augmentation pipeline does (Section 4.4.3). Wu et al. (2020b) identify a specific failure mode of resampling in the *multi-label* setting that a single-label class-balancing analysis misses entirely: oversampling an instance to help its rare label inevitably oversamples every *other* label that instance happens to carry too, so a turn duplicated nine times to help a rare strategy also duplicates whatever common strategies co-occur on that turn. Their Distribution-Balanced (DB) Loss corrects for this with a per-instance, per-class reweighting term. This project independently measured the same failure mode on its own corpus (Section 5.7.3) – effective rebalancing of $6\times$ against an intended $9\times$ – before implementing a DB-style correction, which is reported in the ablation study as one of the project’s controlled experiments rather than assumed to transfer from the vision literature unmodified.

2.4 Transformer Encoders for Dialogue Understanding

The classifier ensemble (Section 4.4.4) draws on four pretrained encoder families: BERT-family masked language models (Devlin et al., 2019), RoBERTa (Liu et al., 2019) at both base and large scale, ELECTRA (Clark et al., 2020), whose replaced-token-detection pretraining objective was hypothesized to transfer well to a discriminative turn-classification task, and TOD-BERT (Wu et al., 2020a), which is pretrained specifically on task-oriented dialogue and was measured to be the one encoder that reliably converts additional dialogue context into predictive signal on this corpus (Section 5.7.2) – a result specific enough to the encoder’s dialogue-pretraining, and not generalizable to the other encoders, that it had to be established empirically rather than assumed.

2.5 Statistical Rigor: Conformal Prediction and Outcome Modeling

Two further bodies of work shape the evaluation methodology (Chapter 5). Conformal prediction (Vovk et al., 2005) provides prediction *sets* with a distribution-free coverage guarantee rather than a single point prediction; this project reports conformal coverage at three risk levels (Section 5.3.4) as a complementary reliability check alongside point-estimate F1. For the donation-outcome analysis, McFadden’s pseudo- R^2 (McFadden, 1974) is the standard goodness-of-fit measure for logistic regression and is used throughout Section 5.6, always reported as a cross-validated out-of-sample estimate rather than the optimistic in-sample figure, since with up to 28 features fitted on 1,017 dialogues the in-sample figure alone would overstate genuine explanatory power.

2.6 Positioning of This Work

Relative to Wang et al. (2019), this project’s contribution is an exhaustive multi-label re-annotation of the full 10,600-turn corpus rather than a sampled single-label subset, and a strategy-to-outcome analysis run on that exhaustive annotation with cross-validated, multiple-testing-corrected statistics rather than raw association tests. Relative to the multi-label classification literature (Ben-Baruch et al., 2021; Wu et al., 2020b), the contribution is empirical rather than methodological: existing techniques (ASL, DB-style rebalancing, empirical-Bayes threshold shrinkage) are

applied and, critically, *measured* on this specific corpus rather than assumed to transfer, and the report documents several cases (Chapter 5) where a technique that is well established elsewhere measurably failed to help, or helped one metric while hurting another, on this dataset – findings that are only visible because of the controlled, sentinel-verified experimental protocol described in Section 4.8.

Chapter 3

Requirements Analysis

3.1 Stakeholders

The system is designed for three kinds of stakeholder. **Researchers** (the project team, and future students building on this corpus) need an annotated dataset and a classifier whose reported numbers can be trusted – i.e., evaluated against an honest ceiling rather than an assumed perfect one. **Nonprofit fundraising practitioners** are the eventual downstream users of the strategy-to-outcome findings: which strategies are worth training callers to use. **Dialogue-system builders** are the users of the donation-intent classifier and its generic-domain extension, who need a model that transfers to a related but distinct persuasive setting rather than one that has memorized this specific corpus.

3.2 Functional Requirements

FR1 – Multi-label strategy taxonomy. The system shall define a fixed, mutually-understood taxonomy of persuasion strategies, organized into higher-level categories, sufficient to describe the range of tactics observed in the corpus.

FR2 – Exhaustive turn-level annotation. The system shall label every persuader turn in the annotated portion of the corpus with the *complete set* of strategies it uses (possibly the empty set), not a single dominant strategy, using the preceding dialogue turns as context where the turn’s meaning depends on it.

FR3 – Automatic strategy prediction. The system shall, given a persuader turn (and up to five preceding turns of dialogue context), predict the probability of each strategy/category being present, and convert these probabilities into a discrete label set via a decision procedure fitted on held-out data.

FR4 – Dual objective support. The system shall support being evaluated and operated at two distinct, independently useful operating points: one that maximizes micro-F1 (overall labeling accuracy, dominated by common strategies) and one that maximizes macro-F1 (equal weight per strategy, sensitive to rare-strategy performance), since a fundraising application interested in *rare but high-value* tactics needs the latter, and one interested in overall labeling throughput needs

the former.

FR5 – Donation-outcome estimation. The system shall, given the (gold) strategy composition of a dialogue, estimate the strategy set’s association with the dialogue’s binary donation outcome, report effect sizes (odds ratios) with significance levels corrected for multiple comparisons, and report the model’s out-of-sample explanatory power.

FR6 – Donation-intent classification. The system shall, given a full persuader–persuadee dialogue, classify whether it ends in a donation intent and, if so, of what type (outright, deferred, or conditional).

FR7 – Cross-domain transfer check. The system’s dialogue-classification architecture shall be evaluated on a second, structurally similar but topically distinct persuasive-dialogue domain to test whether its design generalizes or has overfit to charitable solicitation specifically.

3.3 Non-Functional Requirements

NFR1 – Evaluated against an honest ceiling. Every classification result shall be reported alongside a measured annotation-reliability ceiling (Section 5.2) – how often independent labelings of near-duplicate turns agree with each other – rather than compared only to a theoretical maximum of 1.0, which no annotation process, human or automated, can be expected to reach.

NFR2 – No leakage across the train/validation/test split. Because turns from the same dialogue share context and style, the split into train/validation/test shall be performed at the *dialogue* level, not the turn level, and every downstream statistic (cross-fitted decision thresholds, bootstrap confidence intervals, significance tests) shall resample or fold by dialogue for the same reason.

NFR3 – Rater-agnostic modeling. The classifier shall not be given the identity of the human/LLM annotator who produced a turn’s label as an input feature, even though doing so measurably increases apparent accuracy, because a production system will not know who *would have* labeled an unseen turn, and training on annotator identity would be learning a shortcut rather than the underlying task.

NFR4 – Reproducibility. All randomized procedures (data splitting, weight initialization, data augmentation, cross-fitted decision thresholds, bootstrap resampling) shall use fixed seeds, and a subset of ensemble members (*sentinels*) shall be kept byte-for-byte identical across consecutive design iterations specifically so that any change in their reported score signals a bug rather than a real effect (Section 4.8).

NFR5 – Compute budget. The full training-and-evaluation pipeline shall complete within a single free-tier Kaggle GPU session (an NVIDIA T4×2 instance,

walltime limit on the order of hours, weekly quota limits), and shall be validated on a small data subsample (a *smoke test*) before any full run is committed, to avoid spending scarce GPU-hours on a run that fails partway through from a code defect.

NFR6 – Statistical honesty. Confidence intervals shall be computed by cluster bootstrap over dialogues (not turns), and any comparison between two models or two design choices shall be accompanied by a paired significance test rather than a bare point-estimate difference.

3.4 Data Requirements

The corpus provides 10,600 persuader turns across 1,017 dialogues; the taxonomy requires 41 strategy labels organized into 11 categories (Section 4.2.1); the annotation process is required to report its own inter-annotator (and, where an LLM is involved, inter-run) agreement so that classifier performance can be judged against it rather than in a vacuum (Section 5.2).

3.5 Tooling and Environment Requirements

Model development targets PyTorch with the HuggingFace `transformers` library for encoder fine-tuning, `scikit-learn` for shallow baselines, thresholding, and calibration, and `statsmodels` for the logistic outcome models in Phase 4. Training runs on Kaggle’s free GPU tier; because a single account’s weekly GPU quota is not guaranteed to survive a full seventeen-iteration research process, redundant training across a second Kaggle account was adopted as an infrastructure requirement, not a convenience (Section 4.8).

Chapter 4

System Design and Methodology

4.1 System Overview

The system is organized into four modules built on a shared data foundation (Figure 4.1). **Module A** classifies donation *intent* at the whole-dialogue level. **Module B** – the deepest and most extensively iterated part of this project – recognizes persuasion *strategies* at the turn level, multi-label, at two granularities (41 strategies / 11 categories). **Module C** uses Module B’s gold labels to model donation *outcome*. **Module D** re-applies Module A’s architecture to a second, generic persuasive domain to test transfer. All four modules share the annotated corpus described in Section 4.2.

Table 4.1: End-to-end pipeline, module by module.

Stage	Input	Output	Module
1. Annotation	Raw P4G dialogues (persuader turns)	41-strategy / 11-category multi-label per turn	Data (§4.2)
2. Split	Annotated turns	Dialogue-grouped 70/15/15 train/val/test	Data (§4.2.3)
3. Strategy recognition	Turn text + 5-turn context	Per-label probabilities → decision sets	B (§4.4)
4. Outcome model	Gold strategy sets (dialogue-level)	Strategy → donation association	C (§4.6)
5. Intent classification	Full dialogue transcript	Binary intent + modifier	A (§4.3)
6. Domain transfer	Buyer–seller dialogues	Same intent architecture, new domain	D (§4.7)

4.2 Dataset and Annotation

4.2.1 Taxonomy

The taxonomy was designed to be exhaustive over the tactics observed in a sample of P4G dialogues while staying small enough to annotate and model tractably. It has 11 categories and 41 strategies (Table 4.2), each category holding 3–4 strategies. Each strategy carries a short definition and a small set of example cue phrases used to anchor annotator judgment (and, later, reused as zero-shot cue features for the classifier, Section 4.4.4).

Table 4.2: The 11-category / 41-strategy taxonomy (strategy counts per category).

Category	# Strategies
Rational Appeal	4
Emotional Appeal	4
Credibility Appeal	4
Social Influence	4
Reciprocity and Exchange	3
Commitment and Consistency	4
Framing and Presentation	4
Urgency and Scarcity	3
Threat and Pressure	3
Information Provision	4
Relational and Interactive	4
Total	41

4.2.2 Annotation Process

Annotation was applied to the first 10,600 persuader turns of the corpus (1,017 dialogues), in five contiguous segments produced independently and concatenated in the corpus’s own row order (Table 4.3). Each turn was labeled using its own text plus the five immediately preceding dialogue turns (alternating persuader/persuadee) as context, since several strategies – e.g. a direct answer to a persuadee’s question – cannot be correctly identified from the turn text alone.

Table 4.3: Annotation segments, in canonical corpus row order.

Turn range	Annotator	Turns
1 – 2,000	Segment 1	2,000
2,001 – 4,000	Segment 2	2,000
4,001 – 6,000	Segment 3 (model-batch run)	2,000
6,001 – 7,000	Segment 2 (continued)	1,000
7,001 – 10,600	Segment 4	3,600
Total		10,600

Labeling was carried out by an LLM agent (Claude) working directly from the taxonomy definitions, cue phrases, and decision rules, under a documented **human verification checkpoint**: annotation stopped after an initial batch for manual review against the guideline before the remainder of a segment was processed, and batches were regenerated and re-reviewed where the checkpoint surfaced systematic disagreement with the intended taxonomy semantics. This is disclosed here in full because it directly affects how the classifier’s results should be read: Section 5.2 measures the reliability of *this specific* annotation process (rather than assuming human-level reliability by default) and reports classifier performance against that measured ceiling.

4.2.3 Split

The corpus is split 70/15/15 into train/validation/test using a dialogue-*grouped* stratified split (`MultilabelStratifiedKFold`, 7 folds, fixed seed), so that no dialogue’s turns appear in more than one split – turns from the same dialogue share style and context and would leak information across a turn-level split. This yields 7,592 training, 1,513 validation, and 1,495 test turns. Every downstream cross-validated procedure in this report (decision-threshold fitting, bootstrap resampling, significance testing) resamples by dialogue for the same reason.

4.2.4 Exploratory Statistics

Table 4.4 summarizes the annotated corpus. The mean of 1.68 strategies (1.50 categories) per turn, with fewer than 7% of turns carrying no label at all, confirms that a single-label formulation would misrepresent the data: a persuader turn routinely does more than one persuasive thing at once. Label prevalence is heavily right-skewed at the strategy level (the most common strategy, *rapport_building*, occurs 2,860 times; the rarest, *door_in_the_face*, occurs twice) – an imbalance

ratio of roughly $1,430\times$ – but far more moderate at the category level (*Relational and Interactive* at 46.2% down to *Urgency and Scarcity* at 1.5%, a ratio of roughly $30\times$). This gap between the two granularities’ imbalance is the empirical basis for the category-first reframing described in Section 4.5.

Table 4.4: Corpus statistics (all 10,600 annotated turns).

Turns / Dialogues	10,600 / 1,017
Mean turns per dialogue	10.42 (range 4–15)
Mean strategy-labels per turn	1.679
Mean category-labels per turn	1.504
Turns with zero labels	707 (6.7%)
Turns with ≥ 2 labels	48.9%
Strategy imbalance ratio (max/min prevalence)	$\approx 1,430\times$
Category imbalance ratio (max/min prevalence)	$\approx 30\times$

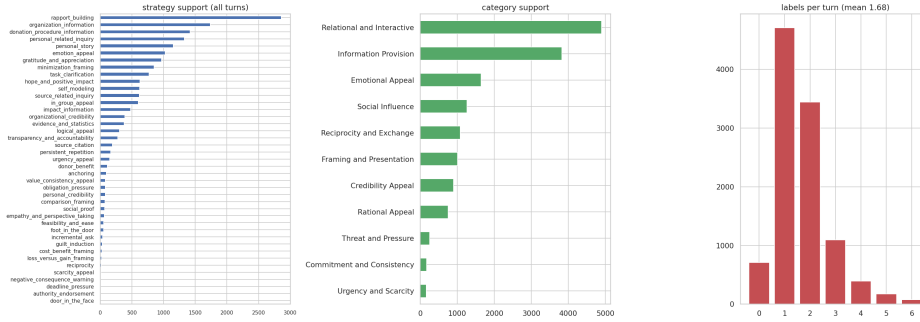


Figure 4.1: Strategy and category label-prevalence distribution across the annotated corpus.

4.3 Module A: Donation-Intent Classification

Module A classifies a complete dialogue’s donation intent (binary: intent present or absent) and, when present, its modifier (outright, deferred, or conditional). Its final architecture – referred to in project records as *Antiphony* (v14) – encodes the persuader’s and persuadee’s turns as two role-pure streams, applies directional cross-attention between the two streams so that each side’s representation is conditioned on the other’s history, and tracks the dialogue’s evolving state with a trajectory recurrent unit before a final classification head. This architecture was reached after thirteen prior iterations spanning flat sequence baselines, speaker-aware encoding, hierarchical dialogue encoders, and specialist routing, each evaluated on held-out data before being superseded (results in Section 5.8).

4.4 Module B: Multi-Label Persuasion-Strategy Classifier

4.4.1 Encoder Architecture

Each ensemble member is a single pretrained transformer encoder with a speaker-role embedding added to its token representations (marking whether each token belongs to a persuader or persuadee turn, resolved from the literal [Persuader]/[Persuadee] markers already present in the input text via the tokenizer’s character-offset mapping), an attention-pooling layer over the resulting sequence (measured to consistently outperform mean pooling), multi-sample dropout before the classification heads for a cheap ensembling-within-a-model effect, and *two* linear heads sharing the pooled representation: a 41-way strategy head and an 11-way category head. Table 4.5 lists the seven encoder members used in the final iteration.

Table 4.5: Encoder ensemble members, final iteration (v17). “Orientation” marks a member as tuned for the micro or macro objective (Section 4.5.3).

Member	Backbone	Context	Orientation
robertaL-ctx0	RoBERTa-large	0 turns	micro
roberta-ctx0	RoBERTa-base	0 turns	micro
electra-ctx0	ELECTRA-base	0 turns	micro
roberta-ctx0-s7	RoBERTa-base	0 turns	macro
roberta-ctx3-mac	RoBERTa-base	3 turns	macro
robertaL-ctx0-mac	RoBERTa-large	0 turns	macro
roberta-ctx0-mac	RoBERTa-base	0 turns	macro

4.4.2 Loss Function

Each head is trained with Asymmetric Loss (Ben-Baruch et al., 2021), which applies a stronger focusing exponent to easy negatives than to positives and subtracts a small probability margin from the negative branch before the log term, so that a label the model is already confidently (and correctly) predicting absent contributes almost nothing to the gradient – important here because with 41 labels the overwhelming majority of (turn, label) pairs are true negatives. *Micro*-oriented members additionally apply Distribution-Balanced rebalancing (Wu et al., 2020b), and *macro*-oriented members apply a per-label positive-class weight (capped to avoid destabilizing training on the rarest labels); the choice of which members get

which treatment is itself an empirical finding, not a default, and is documented in Section 4.5.3.

4.4.3 Data Augmentation

Turns carrying a rare strategy are duplicated with light EDA-style perturbation (synonym replacement and word swap, confined to the turn being labeled so the dialogue context is never corrupted) to give the encoder more gradient signal on the rare tail. This was measured to help some configurations and hurt others (Section 5.7.3), and in the final design is applied only to micro-oriented members.

4.4.4 Ensemble and Decoding

Seven fine-tuned encoders, seven corresponding few-shot nearest-centroid prototype members (a lightweight alternative that scores a turn by cosine similarity to the centroid of each label’s training-turn embeddings, useful as a diversity source for the ensemble search), two shallow TF-IDF classifiers (logistic regression and gradient-boosted trees), and two zero-shot “cue” members scored purely from the taxonomy’s own definition and example-phrase text – 19 base members in all – feed a search over 63 ensembling strategies (simple/weighted means, greedy and bagged-greedy forward selection, tiered support-weighted blends, power-mean pooling, and a level-2 logistic/XGBoost stacker fitted in “long format”, one row per (turn, label) cell, with repeated grouped out-of-fold partitions to avoid the stacker training on rows it will later be scored on).

For every candidate, probabilities are converted to discrete label sets by one of 11 decision rules (Table 4.6), fitted separately for the micro-F1 and macro-F1 objectives by grouped, repeated cross-fitting on the validation split (never the test split). Two of these rules – **eb_shrunk** and **eb_prevalence** – are contributed by this project: each shrinks a label’s own-data threshold toward a pooled anchor with weight $n_j/(n_j + k)$, where n_j is that label’s positive count and k is a single fitted shrinkage constant, so a label with only a handful of positives is pulled almost entirely toward the pooled cut while a well-supported label keeps its own. This is a two-parameter rule, in contrast to the 41-parameter “fit each label’s own threshold independently” rule, and was measured to out-rank that 41-parameter rule on held-out validation data (Section 5.5) – and was in fact selected by the real ensembling search in the final iteration (Section 5.3).

Table 4.6: Decision rules available to the threshold search (11 total).

Rule	Params	Idea
fixed@0.5	0	Flat 0.5 cut, no fitting
global_micro	1	One threshold, tuned for micro-F1
cardinality	1	Threshold set to match mean labels/turn
prevalence	1	Per-label cut anchored to train prevalence
coord_shrunk	≈ 21	Coordinate ascent, shrunk to the global cut
perlabel_shrunk	≈ 21	Own-F1 optimum, shrunk 50% to the global cut
coord_micro / coord_keeppall	41	Unshrunk coordinate ascent variants
perlabel_ownF1	41	Each label’s own-F1-optimal threshold
eb_shrunk*	2	Empirical-Bayes shrinkage to the global cut
eb_prevalence*	2	Empirical-Bayes shrinkage to the prevalence cut

*Contributed by this project.

An explicit parameter-count penalty (a form of model-selection regularization) is applied when comparing rules under the macro objective, so a 41-parameter rule must beat a 1- or 2-parameter rule by a real margin on held-out data rather than by fitting noise in a validation split with as few as two positive examples for the rarest labels.

4.5 The Category-First Reframing

4.5.1 Rationale

Iterations v1–v14 treated the 41-strategy task as primary and read the 11-category view off it as a derived quantity (either via a separate, lightly-weighted auxiliary head or by collapsing strategy predictions onto their parent category). Section 4.2.4 shows why this leaves performance on the table: the category task has a $\sim 30\times$ imbalance ratio against the strategy task’s $\sim 1,430\times$, every one of its 11 labels has non-trivial support in every split (two of the 41 strategies have *zero* test-split positives), and the corpus’s own annotation-reliability ceiling is measurably higher

at the category level than the strategy level (Section 5.2) – meaning there is genuinely more learnable signal left to capture.

4.5.2 Implementation: a Binding Swap

Rather than rewriting the roughly 160 places in the codebase that reference the strategy/category label matrices, the two label sets are bound to two generic *primary/secondary* slots immediately after the label matrices are constructed. Every downstream computation – the loss, the per-label positive-class weights, the ensemble search, both decision-rule searches, every reported metric – is derived from the primary slot, so re-pointing that one binding makes the 11 categories the model’s actual optimization target everywhere at once, with no change to any algorithm. This let the same, already-validated pipeline code serve both configurations and made every resulting bug (a few genuinely surfaced, entirely in code that had implicitly assumed which of the two label sets was primary, e.g. a hierarchical-gating step that multiplies a label’s probability by its parent category’s probability, which has no meaning once the category *is* the top-level label) a loud crash caught by the smoke-testing protocol (Section 4.8) rather than a silent wrong number.

4.5.3 The Objective Split

A second empirical finding shapes the final member pool. Two class-imbalance interventions – Distribution-Balanced rebalancing and EDA augmentation – were each measured, in controlled single-member experiments against a byte-identical baseline, to move micro-F1 and macro-F1 in *opposite* directions (Table 5.11). Rather than picking one setting for the whole pool, the final design gives each objective its own members: micro-oriented members keep augmentation on and add DB-rebalancing; macro-oriented members turn augmentation off and rely on positive-class weighting instead. This mirrors, and extends to two additional interventions, a principle already established earlier in the project’s history for the positive-weighted loss term itself.

4.6 Module C: Donation-Outcome Analysis

Module C fits a sequence of nested logistic regressions on the 1,017 dialogues, using *gold* (human/LLM-annotated) strategy labels exclusively – never the classifier’s own predictions – specifically to avoid contaminating the outcome analysis with the classifier’s own errors (the classifier was trained on 70% of these same dialogues, so

its predictions on them are optimistic). Four models are compared: **M1** uses only the 11 category-presence indicators; **M2** adds two sentiment/engagement proxy features; **M3** isolates the guilt-induction $\times$ reciprocity interaction the project’s own prior working notes flagged for a replication check; **M4** adds the top co-occurring strategy pairs and dialogue-level richness features (distinct-strategy count, distinct-category count, total strategy instances). Each model’s fit is reported both in-sample and via 5-fold cross-validated McFadden pseudo- R^2 , since with up to 28 features on 1,017 dialogues the in-sample figure alone materially overstates genuine out-of-sample explanatory power.

A single-label *projection* of the same gold data (keeping only each turn’s most-frequent, rarest, or a random one of its co-occurring labels, averaged over five seeds) is fit under the identical M1–M4 specification, to isolate whether allowing a turn to carry multiple simultaneous strategies is itself adding predictive value, independent of any other difference between this project’s corpus and prior single-label work (Section 5.6).

The team’s own working notes on this module (retained in the project archive) flag a limitation given prominent treatment in Chapter 5 rather than left implicit: several of the strongest strategy–donation associations are measured on a bag-of-strategies representation that discards *when* in the dialogue a strategy occurred, and a strategy that predominantly occurs *after* a persuadee has already committed (e.g. expressing gratitude) will show a strong positive association that is a consequence of the donation, not a cause of it. This project reports that limitation explicitly as a scope boundary of the current outcome model, rather than resolving it, and names the temporal audit required to resolve it as future work (Section 7.3).

4.7 Module D: Generic Cross-Domain Extension

To test whether Module A’s dual-stream cross-attention architecture is specific to the donation domain or genuinely general to persuasive two-party dialogue, the same architecture was retrained (from the same code path, no domain-specific changes) on a purpose-built 500-dialogue, 7,953-turn corpus of English buyer–seller price-negotiation dialogues (40 products across 14 categories, 10 synthetic buyer archetypes, generated and then LLM-rewritten for naturalness, with an explicit English-only validation pass). Results are reported in Section 5.9.

4.8 Experimental Methodology and Reproducibility Engineering

Module B’s seventeen-iteration history is itself a designed part of the project’s methodology, not an incidental record of trial and error. Each iteration follows the same protocol:

1. **A stated hypothesis.** Every change is motivated by a specific, falsifiable claim about what should happen to which metric, derived from either the classification literature (Chapter 2) or a measurement taken on this project’s own prior results.
2. **Sentinel members.** At least two ensemble members carry no new configuration flags between an iteration and the previous one. Their reported score must reproduce to four decimal places (it did, in every iteration reported in this chapter, for every member trained at the base 128-token sequence length; members trained at longer sequence lengths were measured to carry a small non-determinism floor of about ± 0.005 , attributable to a GPU non-deterministic kernel path, and are treated as the noise floor below which no effect is claimed as real). A sentinel that moves invalidates the whole comparison and is treated as a bug, not a result.
3. **A GPU-free smoke test first.** A local harness re-executes every notebook section against the real corpus with synthetic (but support-calibrated) transformer outputs standing in for the GPU-trained members, exercising every downstream code path – the ensemble search, both decode searches, the significance tests, Module C – in well under a minute. Only once this passes does an actual GPU run begin, and only on a small data subsample (a *smoke test* in the GPU-run sense) before the full multi-hour run is committed.
4. **Redundant infrastructure.** Because Kaggle enforces a weekly per-account GPU quota that a single iterative research process can and did exhaust, every full run was queued on two independent Kaggle accounts in parallel, both smoke-tested first; this additionally gave, at several points in the project, an independent reproducibility check on the same configuration run twice.
5. **A recorded verdict.** Every change is explicitly marked adopted, rejected, or partially adopted based on what the held-out numbers actually showed, and rejected ideas are kept, commented, in the codebase rather than silently deleted, so a plausible-sounding idea that already failed is not re-attempted. Table 5.11 in Chapter 5 is the consolidated record of every such verdict.

Chapter 5

Evaluation and Results

5.1 Evaluation Protocol

All results in this chapter are computed on the held-out test split (1,495 turns / 152 dialogues, Section 4.2.3), which no model selection, threshold fitting, or hyperparameter decision ever touched. Confidence intervals are cluster bootstraps over *dialogues* (2,000 resamples); significance tests are paired, also resampled by dialogue. Macro-F1 is reported over the labels actually *present* in the test split – a label with zero test positives has an undefined F1 (a 0/0 recall), and averaging it in as a hard zero mechanically caps the score before any model has done anything wrong. Two of the 41 strategies (*authority_endorsement*, *door_in_the_face*) have zero test-split positives, purely from their extreme corpus rarity (5 and 2 occurrences in the whole 10,600-turn corpus); every category has non-zero test support. Where earlier design iterations reported the old all-label-averaged figure, this chapter converts it for direct comparability and states so explicitly.

5.2 Annotation-Reliability Ceiling

Before any classifier result is judged, the annotation process itself is measured, because no classifier trained on an imperfect label source can honestly be expected to exceed the reliability of that source. This is estimated from *near-duplicate* turns – turns with identical or near-identical text that were independently labeled more than once (either by the same annotation run recurring, or, where corpus text repeats naturally, by different segments) – via the agreement F1 between their two label sets, converted to a *consensus-predictor bound* $2q/(1+q)$, the score a hypothetical predictor of the pair’s consensus label would achieve against either individual labeling.

Table 5.1: Annotation-reliability ceiling, strategy level (41-way) vs. category level (11-way).

	Strategy (41)	Category (11)
Same-annotator agreement (near-dup.)	0.805	0.871
Cross-annotator agreement (near-dup.)	0.578	0.657
Consensus bound, self	0.892	0.931
Applicable bound (model is rater-agnostic)	0.733	0.793

Because the classifier is deliberately rater-agnostic (NFR3, Section 3) – it is never told which annotator produced a label, since a deployed system would not know this for a new turn – the relevant ceiling is the *cross*-annotator consensus bound, not the higher same-annotator figure. This is 0.733 at the strategy level and 0.793 at the category level: a real, measured gap in favor of the coarser granularity, and part of the evidence base for the category-first reframing (Section 4.5).

5.3 Category-Level Results (Primary Task)

5.3.1 Headline Results

The final system (v17) reaches **micro-F1 0.706**, **macro-F1 0.564** at its micro-optimal operating point, and **micro-F1 0.692**, **macro-F1 0.559** at its macro-optimal operating point (Table 5.2). This is, on both metrics simultaneously, the best result of any of the seventeen design iterations: every earlier iteration had to trade one metric for the other at a given operating point (Table 5.3). A micro-F1 of 0.706 sits above the corpus’s own *cross-annotator* human agreement of 0.657 on the same task – the model already agrees with a consensus label about as often as two independent human annotations agree with each other – and within 9 points of the 0.793 theoretical ceiling.

Table 5.2: Final system (v17), category task, both operating points, test split. 95% CIs are cluster bootstrap over dialogues.

Operating point	Micro-F1	95% CI	Macro-F1	95% CI
Micro-optimal	0.706	[0.685, 0.727]	0.564	[0.533, 0.589]
Macro-optimal	0.692	[0.671, 0.712]	0.559	[0.535, 0.580]
<i>Gold cardinality (mean labels/turn, test)</i> 1.56				

Table 5.3: Category-task headline progression across design iterations (test split; all figures reported on the present-label macro-F1 definition for comparability). “Config” summarizes the governing design decision of that iteration.

Iter.	Config	Point	Micro-F1	Macro-F1	Card.
v10	Category as light auxiliary ($w=0.3$)	micro	0.707	0.540	1.65
v10	Category as light auxiliary ($w=0.3$)	macro	0.697	0.560	1.56
v14	Category as light auxiliary ($w=0.3$), pruned pool	macro	0.687	0.556	1.90
v15	Category made <i>primary</i> ($w=1.0/0.3$)	micro	0.701	0.540	1.71
v16	Both heads equal weight ($w=1.0/1.0$)	macro	0.689	0.560	1.86
v17	+ macro-oriented pool, new de- code rules	micro	0.706	0.564	1.71

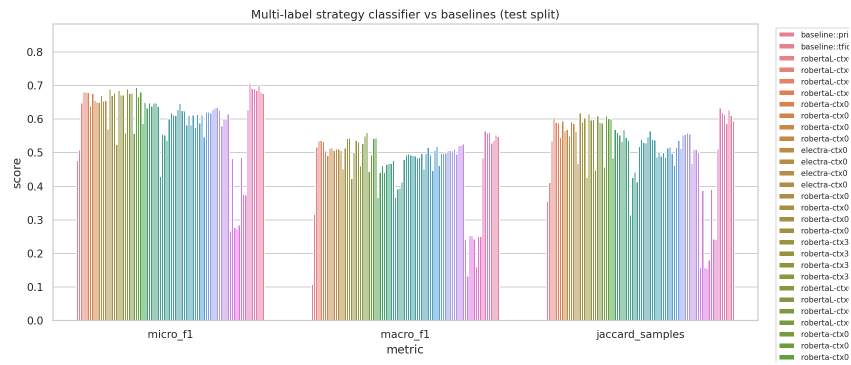


Figure 5.1: Micro-F1 across baselines and the final ensemble, category task.

5.3.2 Per-Category Breakdown and the Rare-Category Gap

Table 5.4: Per-category precision/recall/F1, final system (v17), macro-optimal point, test split.

Category	Support	Precision	Recall	F1
Commitment and Consistency	22	0.147	0.227	0.179
Urgency and Scarcity	23	0.146	0.261	0.188
Threat and Pressure	33	0.220	0.394	0.283
Rational Appeal	104	0.459	0.692	0.552
Credibility Appeal	131	0.717	0.695	0.705
Reciprocity and Exchange	146	0.901	0.808	0.852
Framing and Presentation	158	0.712	0.734	0.723
Social Influence	193	0.492	0.487	0.490
Emotional Appeal	248	0.609	0.766	0.679
Information Provision	569	0.694	0.861	0.769
Relational and Interactive	701	0.690	0.786	0.735

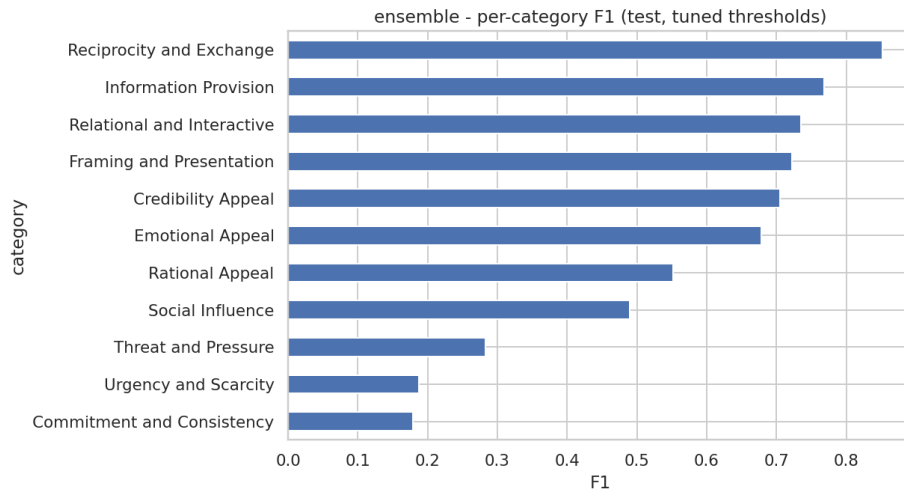


Figure 5.2: Per-category F1, sorted by support – the three rarest categories are the visible outliers.

Table 5.4 makes the shape of the remaining error obvious: the three rarest categories (3.4% of all positive labels in the test split) score a mean F1 of 0.22, against 0.69 for the other eight. If those three matched the other eight, overall macro-F1 would be approximately **0.68** rather than 0.56 – essentially the entire remaining gap to a near-ceiling result is concentrated in these three labels.

This gap is *not* monotonically closing across iterations, and this report states

that plainly rather than only reporting the headline number that improved. Table 5.5 tracks the rare-category mean F1 against the other eight categories’ mean F1 across every category-primary iteration: the equal-weight multi-task change (v15→v16) improved the rare categories specifically (the mechanism it was intended to trigger, Section 5.7.4), but the subsequent pool change (v16→v17) improved the *overall* headline by improving the common eight categories further while the rare three regressed slightly. The best-ever headline result in Table 5.2 is therefore not evidence that the rare-category problem is solved; it is evidence that a different lever (encoder pool composition) helps a different part of the label space, and the rare-category gap remains the right target for future work (Section 7.3).

Table 5.5: Rare-category (3 rarest) vs. common-category (other 8) mean F1 across iterations, macro-optimal point.

Iteration	Rare-3 mean F1	Other-8 mean F1	Overall macro
v14 (category auxiliary)	0.184	0.662	0.556
v15 (category primary)	0.201	0.680	0.549
v16 (equal-weight multi-task)	0.237	0.681	0.560
v17 (+ macro-oriented pool)	0.216	0.688	0.559

5.3.3 Statistical Significance

Table 5.6: Paired significance tests, cluster bootstrap over dialogues, final system (v17).

Comparison	Point	Metric	Δ	95% CI	Sig.
Ensemble vs. best single encoder	micro	micro-F1	+0.025	[0.015, 0.034]	✓
Ensemble vs. best single encoder	macro	macro-F1	+0.024	[−0.002, 0.049]	× ($p=0.069$)
Ensemble vs. TF-IDF base-line	micro	micro-F1	+0.057	[0.042, 0.072]	✓
Ensemble vs. TF-IDF base-line	macro	macro-F1	+0.090	[0.065, 0.115]	✓
micro-opt vs. macro-opt point	–	micro-F1	+0.015	[0.008, 0.021]	✓
micro-opt vs. macro-opt point	–	macro-F1	+0.004	[−0.015, 0.022]	× ($p=0.664$)

Two results in Table 5.6 are reported precisely because they complicate the headline story and should not be glossed over. First, the ensemble’s macro-F1 advantage

over its single best encoder member is *not* statistically significant ($p = 0.069$) – the ensembling machinery (Section 4.4.4) earns a clearly significant micro-F1 gain and a significant gain over the non-transformer baseline on both metrics, but its edge over one well-configured transformer on macro-F1 specifically should be described as suggestive, not established. Second, the micro-optimal decoding point’s macro-F1 is statistically indistinguishable from the macro-optimal point’s macro-F1 ($p = 0.664$); part of Table 5.2’s headline macro-F1 gain over earlier iterations reflects which candidate a stochastic ensembling search happened to select, not a real improvement in what the underlying probabilities encode (Section 5.7.1).

5.3.4 Conformal Prediction Sets

As a complementary, distribution-free reliability check, split conformal prediction (Vovk et al., 2005) is applied at three target coverage levels (Table 5.7). At 80% target coverage, empirical coverage of 0.789 is achieved with a mean prediction-set size of 2.67 labels (against a true mean cardinality of 1.56) – the calibration is close to nominal, and the set-size cost of that calibration is a further, complementary view of how much genuine per-turn ambiguity the model is carrying, beyond what a single point-estimate F1 conveys.

Table 5.7: Split-conformal prediction coverage, category task, test split.

Target coverage	Empirical coverage	Mean set size	Micro-F1	Macro-F1
0.90	0.878	3.75	0.526	0.454
0.80	0.789	2.67	0.587	0.509
0.70	0.693	2.00	0.626	0.536

5.4 Strategy-Level (41-Way) Results

Before the category-first reframing, the strategy-primary line of iterations peaked at **micro-F1 0.579**, **macro-F1 0.350** (micro-optimal) and **micro-F1 0.563**, **macro-F1 0.359** (macro-optimal) at iteration v13 – the best result achieved under that framing, itself reached by *removing* four measurably harmful or dominated ensemble members rather than adding anything (Section 5.7). This is retained in this report both because Module B’s 41-strategy head remains a live output of the final system (as the secondary/auxiliary head, Section 4.5) and because the comparison against Table 5.2 is the direct empirical justification for why the category task became primary: at its best, the strategy-level macro-F1 (0.359) sits at only 49% of its own applicable ceiling (0.733), where the category task’s final

macro-F1 (0.564) sits at 71% of its ceiling (0.793) – the category task simply has substantially more learnable headroom.

5.5 Decision-Rule Study

Which of the eleven decision rules (Table 4.6) should convert probabilities to label sets is itself fitted on held-out data, never on the test split. Table 5.8 reports every rule’s mean macro-F1 under repeated grouped cross-fitting on the validation split (4 repeats $\times$ 4 folds, category task, v16 probabilities): the two empirical-Bayes rules this project contributes rank first and second, narrowly ahead of the 41-parameter own-threshold rule and comfortably ahead of the single global threshold and prevalence-anchored rules. Confirmed against the (never-fitted-to) test split, the same relative ordering holds, which is itself worth reporting: on the strategy task, an earlier design iteration found that a 41-parameter rule’s validation-set ranking did *not* reliably predict its test-set behavior (a symptom of overfitting a validation split as small as 1,513 rows), whereas on the better-supported category task the validation ranking is a trustworthy guide to test performance.

Table 5.8: Decision-rule study, category task, held-out validation (repeated grouped cross-fitting). Rules marked * are contributed by this project.

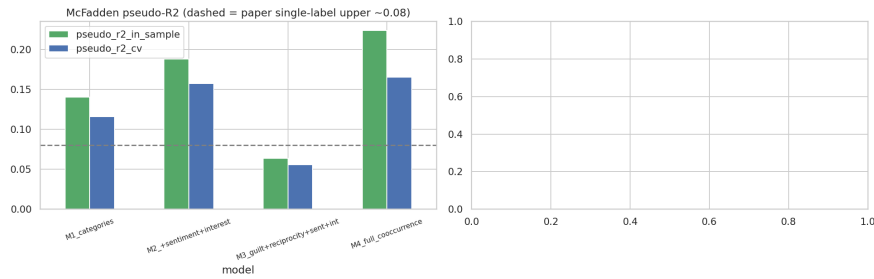
Rule	Val. macro-F1	Std. dev.	Params
eb_shrunk*	0.5646	0.0174	2
eb_prevalence*	0.5645	0.0142	2
perlabel_ownF1	0.5642	0.0154	41
perlabel_shrunk	0.5551	0.0173	21
prevalence	0.5431	0.0174	1
global_micro (best micro-F1: 0.700)	0.5249	0.0210	1

5.6 Donation-Outcome Analysis (Module C)

Module C’s logistic outcome models are fit on the 1,017 annotated dialogues’ *gold* strategy labels. Table 5.9 reports each model’s cross-validated pseudo- R^2 alongside its in-sample figure, which is consistently and substantially higher – a direct illustration of why the out-of-sample figure, not the in-sample one, is the number to trust with up to 28 features fit on 1,017 observations.

Table 5.9: Donation-outcome logistic models (McFadden pseudo- R^2), gold labels.

Model	Features	In-sample R^2	CV R^2	CV std.
M1: 11 categories	11	0.140	0.116	0.036
M2: M1 + sentiment/interest	13	0.188	0.157	0.038
M3: guilt $\times$ reciprocity + sent./interest	4	0.064	0.056	0.024
M4: M1 + co-occurrence + richness	28	0.224	0.165	0.046

**Figure 5.3:** Donation-outcome model diagnostics (Module C).

M4’s cross-validated pseudo- R^2 of 0.165 is roughly double the 0.015–0.08 range the team’s prior working notes recorded for a single-label specification of a related outcome-modeling task (Team internal working notes – source to be confirmed, 2026). This project treats that specific comparison with caution and states the reason explicitly: the two settings differ in labeling scheme, in the outcome variable definition, *and* in feature count simultaneously, so the improvement cannot be attributed to multi-labeling alone from this comparison. To isolate the multi-labeling question specifically, a single-label *projection* of this project’s own gold data is fit under the identical M1–M4 specification, holding the corpus, the outcome definition, and the feature-construction code fixed and changing only whether a turn may carry one label or several. Under this controlled comparison, multi-label M4 reaches a cross-validated pseudo- R^2 of 0.165 against 0.133–0.138 for the best single-label projection (commonest-label, rarest-label, and five-seed-random-label projection rules), a genuine **+0.03** advantage for multi-labeling that survives every projection rule tried – the honest, narrowly-scoped version of the multi-label claim, in contrast to the roughly +0.15 that the uncontrolled cross-paper comparison would suggest.

5.6.1 Category–Donation Associations

Table 5.10: Category odds ratios for donation outcome, M1 (gold labels, significant at $p < 0.01$ in bold).

Category	Coefficient	Odds Ratio	p -value
Reciprocity and Exchange	1.575	4.83	<0.001
Threat and Pressure	−0.927	0.40	<0.001
Information Provision	1.082	2.95	0.001
Emotional Appeal	0.231	1.26	0.183
Social Influence	0.208	1.23	0.200
Framing and Presentation	0.147	1.16	0.362
Credibility Appeal	0.105	1.11	0.511
Urgency and Scarcity	0.143	1.15	0.584
Relational and Interactive	−0.191	0.83	0.651
Rational Appeal	0.030	1.03	0.854
Commitment and Consistency	0.008	1.01	0.972

Dialogues that use reciprocity-based appeals (e.g. emphasizing the mutual benefit of donating) have nearly $5\times$ the odds of ending in a donation, controlling for the other ten categories; dialogues that lean on threat-and-pressure tactics have well under half the odds. Two further richness measures confirm the same direction with non-parametric tests: a Mann–Whitney comparison of donated vs. non-donated dialogues finds significantly more distinct strategies (mean 9.64 vs. 8.56, $p < 0.001$), more distinct categories (5.97 vs. 5.33, $p < 0.001$), and more total strategy instances (17.95 vs. 16.32, $p < 0.001$) in dialogues that end in a donation. A prior single-label analysis of a related corpus reported a mean of only 4–5 distinct strategies per dialogue (Team internal working notes – source to be confirmed, 2026); this project’s multi-label annotation recovers a mean of **9.34**, a direct consequence of allowing a turn to carry more than one simultaneous label rather than a difference in persuader behavior.

5.6.2 A Limitation Stated Explicitly: Temporal Direction

The odds ratios in Table 5.10 are estimated from a bag-of-strategies representation of each dialogue and therefore cannot distinguish a strategy that *causes* donation from a strategy that is a *reaction to* an already-forming decision. This project’s own working notes flag a concrete instance: gratitude-expressing strategies occur disproportionately late in a dialogue, and a preliminary position-windowed check

found that roughly two-thirds of a large raw gratitude-donation association in earlier project notes disappeared once only gratitude expressed *before* the point where the persuadee could plausibly have already committed was counted – consistent with a persuader thanking a donor *after* they agree, rather than gratitude persuading them to agree. This project reports that finding as a scope boundary rather than resolving it here: the position-windowed check used a quick, single-strategy-at-a-time comparison with small per-window sample sizes and is not presented in Table 5.10 or treated as a validated result. The temporal, commitment-turn-aware hazard-model analysis needed to resolve it rigorously is named as future work (Section 7.3) rather than included here, precisely so that Table 5.10 is not overstated as more causally interpretable than an observational, un-randomized corpus of this size can actually support.

5.7 Ablation Study

Table 5.11 consolidates every controlled intervention tested across the project’s seventeen design iterations. Each row is a single, isolated change measured against a sentinel-verified baseline (Section 4.8); “Verdict” records whether the change was kept in the final system.

Table 5.11: Consolidated ablation record. Δ_{mi} / Δ_{ma} are the measured change in micro-F1 / macro-F1 for that specific, isolated intervention.

Intervention		Task	Member(s)	Δ_{mi}	Δ_{ma}	Verdict
Target-span pooling	attention	Strategy	ctx1 pair	+0.002	−0.007	Rejected (underpowered)
Target-span retest	pooling,	Strategy	ctx3 pair	−0.008	+0.010	Rejected (a trade)
Prior-initialized bias	head	Strategy	1 member	−0.008	−0.012	Rejected (harmful)
Context (dialogue-pretrained enc.)	width 0→3	Strategy	TOD-BERT	+0.028	+0.017	Adopted
Context (general enc.)	width 0→3	Strategy	RoBERTa	−0.005	+0.022	Adopted
Context architecture)	width 0→3 (3rd architecture)	Strategy	ELECTRA	−0.001	−0.039	Rejected (architecture-specific)
DB-rebalanced weighting	loss	Strategy	ELECTRA	+0.009	−0.007	Kept, micro-members only
EDA on/off	augmentation	Strategy	RoBERTa	+0.014	−0.009	Split, macro-members off
Pruning blend weight	by ensemble-	−	−	n/a	n/a	Rejected (weights not reproducible)
Category task	as primary	Category	full pool	+0.001	−0.007	Neutral alone
Equal-weight loss	multi-task	Category	full pool	+0.001	+0.009	Adopted
Context	width 0→3	Category	4-member set	+0.005	−0.005	Rejected (no macro gain)
Macro-oriented expansion	pool ex-	Category	+2 members	+0.006	+0.018	Adopted (best-ever)

5.7.1 Selection Noise: a Methodological Caution

A negative-space finding is reported here on equal footing with the positive ones, because it changed how every later iteration was evaluated. Running an identical configuration twice (on two independent Kaggle accounts, Section 4.8) produced ensemble-blend weights that differed by up to $2.7\times$ between the two runs for the same member, with one member receiving zero weight in one run and non-trivial weight in the other. The stochastic bagged-greedy search that selects the final ensemble blend is simply not a reproducible enough quantity to prune members by: doing so would be pruning on noise, the same failure mode this project guards

against when selecting a decision rule (Section 5.5). Every pruning decision in Table 5.11 is instead based on deterministic *member-level* scores, which reproduce to four decimal places (Section 4.8).

5.7.2 Context Width: an Architecture-Dependent Effect

Whether giving an encoder more preceding dialogue turns as context helps is not a single, corpus-wide answer – it depends on the encoder’s own pretraining, and had to be measured per architecture rather than assumed. Extending the context window from 0 to 3 preceding turns (strategy task, Table 5.11) helped TOD-BERT substantially on both metrics (+0.028 micro / +0.017 macro) and helped RoBERTa on macro-F1 specifically (−0.005 micro / +0.022 macro), but measurably *hurt* ELECTRA (−0.001 micro / −0.039 macro). TOD-BERT is pretrained specifically on task-oriented dialogue (Wu et al., 2020a), which is the most plausible explanation for why it is the one encoder that reliably turns additional context into signal; ELECTRA has no such dialogue-specific pretraining and its discriminative replaced-token objective may be comparatively more sensitive to the added, potentially distracting, context tokens. On the category task specifically, the same intervention (applied to a four-member subset of the pool) was re-measured and found *not* to help macro-F1 at all (−0.005), a task-specific null result reported alongside the strategy-task positive one precisely because the two tasks disagree, and assuming an effect measured on one task transfers to the other would have been an error.

5.7.3 Augmentation and Rebalancing: a Trade, Not a Fix

The EDA rare-class augmentation (Section 4.4.3) was built to help macro-F1 – more copies of a turn carrying a rare strategy should mean more gradient signal for that strategy. Measured in isolation, it does the opposite: +0.014 micro-F1 but −0.009 macro-F1 on the same member. The mechanism, diagnosed directly from the corpus rather than inferred from the result alone: oversampling a turn to help its rare strategy also oversamples whichever *common* strategies happen to co-occur on that same turn (Section 2), so a strategy intended to receive a $9\times$ boost receives it, but every common co-occurring strategy receives an unintended $\sim 1.5\times$ boost as a side effect – roughly a third of the intended rebalancing is spent inflating labels that were already abundant. Distribution-Balanced rebalanced loss weighting (Wu et al., 2020b), which damps a common label’s loss contribution specifically on turns it is riding alongside a rare label, shows the identical micro-positive/macro-negative pattern (+0.009/−0.007) via a different mechanism (precision recovery on the common labels, rather than more signal for the rare ones). Both findings point the same direction and motivated the final design’s objective-split principle

(Section 4.5.3): each intervention is a genuine trade between the two metrics, so rather than picking one setting for the whole ensemble, micro-oriented members keep augmentation on and add DB-rebalancing, and macro-oriented members turn augmentation off and rely on positive-class weighting instead.

5.7.4 Equal-Weight Multi-Task Loss: Mechanism Check

The equal-weight multi-task change (raising the auxiliary strategy head’s loss weight from 0.3 to 1.0, category task, Table 5.11) is the one intervention in this study whose effect was verified against a specific, falsifiable mechanism rather than reported as a bare number: it was hypothesized to help specifically because the fine-grained 41-strategy gradient should regularize hardest exactly where category-level supervision alone is weakest – the three rarest categories, each built from strategies that are themselves individually rare. Table 5.5 confirms this directly: the rare-3 mean F1 rose from 0.201 to 0.237 while the common-8 mean stayed flat (0.680→0.681) across this specific change, which is the pattern the mechanism predicts and not the pattern a generic regularization effect (which would move both groups together) would produce.

5.8 Module A: Donation-Intent Classification

Table 5.12: Donation-intent classification (Module A), binary task, test split.

Model	Accuracy	Macro-F1
Majority-class baseline	72.5%	0.421
Flat single-stream baseline (earliest iteration)	–	≈0.63
Hierarchical + specialist routing (intermediate)	–	0.834
Dual-stream cross-attention (final, v14-Antiphony)	89.5%	0.862

Module A’s final architecture encodes the persuader’s and persuadee’s turns as separate role-pure streams and applies directional cross-attention between them before a trajectory-tracking recurrent layer (Section 4.3). Every backbone tested (RoBERTa, DeBERTa-v3, TOD-BERT) improved when moved from a flat single-stream encoding to this dual-stream design, indicating the gain is attributable to the architecture rather than to any one pretrained encoder.

5.9 Module D: Cross-Domain Generalization

Table 5.13: Module A’s architecture retrained on the generic buyer–seller negotiation domain (500 dialogues, binary outcome), test split.

Backbone	Accuracy	Macro-F1
Majority-class baseline	53.3%	0.348
TOD-BERT	92.0%	0.920
RoBERTa-base	93.3%	0.933
DeBERTa-v3-base	97.3%	0.973

The same dual-stream architecture, retrained without domain-specific modification on a structurally similar but topically distinct buyer–seller price-negotiation corpus, reaches 97.3% accuracy with its strongest backbone – evidence that the architectural gain measured in Module A (Table 5.12) is a property of the dual-stream design applied to two-party persuasive dialogue in general, and not an artifact specific to the donation-solicitation domain or to this particular corpus.

5.10 Summary

Across all four modules, the results in this chapter support three higher-level conclusions that carry into Chapter 7: (i) evaluating a classifier against its corpus’s own measured annotation ceiling, rather than against 1.0, changes which results should be considered successes – the final category-level micro-F1 of 0.706 is close to that ceiling and should be read accordingly; (ii) a disciplined, hypothesis-driven, sentinel-verified experimental protocol is what makes it possible to tell a real effect (context width for a dialogue-pretrained encoder) from a plausible-sounding one that measurably fails (target-span pooling, prior-initialized biases) or a genuine trade rather than a free improvement (augmentation, class-imbalance reweighting); and (iii) the remaining error is not evenly distributed – it is concentrated in three rare categories that the strongest single lever found so far (equal-weight multi-task supervision) helps, but does not solve, leaving a clearly scoped target for future work.

Chapter 6

Project Management, Tools, and Impact

6.1 Team Organization

The project was organized around the four modules described in Chapter 4, with primary ownership assigned per module while the underlying annotated corpus (Section 4.2) was a shared, jointly-maintained resource that every module depended on.

Table 6.1: Module ownership.

Module	Scope
A – Donation-intent classification	Dual-stream cross-attention architecture (v1–v14), binary intent + modifier classification.
B – Persuasion-strategy classification	Taxonomy design, corpus annotation coordination, the 17-iteration multi-label classifier (v1–v17), the decision-rule and ensembling study.
C – Donation-outcome analysis	Logistic outcome modeling, significance testing, single-label projection comparison, temporal-confound analysis.
D – Generic domain extension	Synthetic buyer–seller corpus construction, cross-domain retraining and evaluation of Module A’s architecture.

6.2 Development Methodology

Module B in particular followed the iterate-under-a-hard-compute-budget protocol detailed in Section 4.8: every one of its seventeen iterations was smoke-tested locally (GPU-free, against the real corpus, in under a minute) and then smoke-tested again on Kaggle (a small data subsample, a few minutes of GPU time) before a full multi-hour training run was committed, and every full run’s results were checked against deterministic sentinel members before being trusted. This caught

real defects before they cost GPU-hours on several occasions – most notably when the category-first reframing (Section 4.5) was implemented, where the local smoke test caught an architectural incompatibility (a hierarchical-gating step that no longer had a well-defined coarser label to gate against) before any Kaggle run was attempted.

6.3 Tools and Infrastructure

Table 6.2: Core tools and infrastructure.

Purpose	Tool
Model training	PyTorch, HuggingFace transformers
Shallow baselines, calibration	scikit-learn
Statistical / out-come modeling	statsmodels
GPU compute	Kaggle (NVIDIA T4×2, free tier, two accounts)
Version control	Git
Notebook generation	A single builder script regenerates each iteration’s notebook, so every change is a reviewable source-code diff rather than a manually edited <code>.ipynb</code>

Kaggle’s free tier enforces a weekly per-account GPU-hour quota, which a seventeen-iteration research process is not guaranteed to fit inside a single account. Every full training run was accordingly queued redundantly across two independent Kaggle accounts, which both protected the project against losing a run to quota exhaustion mid-iteration and, on several iterations, provided an independent reproducibility check by re-running the identical configuration twice.

6.4 Risk Management

Table 6.3: Key risks and mitigations.

Risk	Mitigation
GPU quota exhaustion mid-experiment	Redundant training across two Kaggle accounts (Section 4.8).
A multi-hour GPU run fails from a code defect	Mandatory local (GPU-free) and Kaggle-side (data-subsample) smoke tests before every full run.
A design change looks like an improvement but is measurement noise	Deterministic sentinel ensemble members required to reproduce exactly before any comparison is trusted (Section 4.8); decision rules and pruning decisions fitted only on held-out validation data, never the test split.
Extreme class imbalance makes the rarest labels unlearnable	Measured, rather than assumed: two strategies with zero test-split support are excluded from macro-F1 by definition (Section 5) rather than silently scored as failures; the category-first re-framing directly targets the imbalance that <i>is</i> addressable.
Overclaiming causal or comparative results	Explicit statistical significance testing on every headline comparison (Table 5.6); the single-label projection control (Section 5.6) built specifically to avoid an uncontrolled cross-paper comparison; the temporal-confound limitation stated explicitly rather than left implicit (Section 5.6).

6.5 Ethical Considerations

LLM-assisted annotation. A substantial portion of the strategy annotation was produced by an LLM agent under a human verification checkpoint (Section 4.2), not purely by human annotators. This is disclosed in the declaration, in the annotation methodology section, and here, and its reliability is measured directly (Section 5.2) rather than assumed to match human-only annotation quality; every classifier result in this report is explicitly evaluated against that measured ceiling.

Dual-use of persuasion research. A system that recognizes and quantifies the effectiveness of persuasion strategies is, by construction, also a system that

could be used to make persuasion more manipulative rather than more informative – e.g. to identify which pressure-based tactics (Table 5.10 shows *Threat and Pressure* strategies are associated with a *lower*, not higher, donation rate, which is itself a finding against using such tactics) extract compliance rather than genuine willingness. This project’s scope is limited to a single benign, well-studied domain (charitable solicitation for a specific, named charity) and the associational findings in Chapter 5 are reported with their causal limitations stated explicitly (Section 5.6) specifically so they are not misapplied as a validated manipulation playbook.

Donor and participant privacy. The underlying P4G corpus consists of crowd-worker dialogues released publicly by its original authors (Wang et al., 2019) with identifying information already removed; this project introduces no new personally identifying data and the generic extension corpus (Module D) is entirely synthetic.

Fairness. The classifier is deliberately rater-agnostic (NFR3, Chapter 3): it is never given the identity of the turn’s annotator as a predictive feature, even though doing so was measured to increase apparent in-corpus accuracy, because that would be learning a shortcut specific to this annotation process rather than the underlying persuasive-language task, and would not transfer to a deployment setting where the "annotator" of a new turn does not exist.

6.6 Broader Impact

The strategy taxonomy, the annotated corpus, and the strategy-to-outcome findings (Chapter 5) have direct potential use as fundraiser-coaching material: the odds-ratio analysis (Table 5.10) is, in effect, evidence-based guidance that reciprocity-and-exchange and information-provision framing are associated with better outcomes than threat-and-pressure tactics in this domain, for an organization that wants that guidance grounded in dialogue data rather than intuition. More broadly, the annotated corpus and the released classifier give the wider research community a resource for testing persuasion-theory predictions (Cialdini’s principles among them, Chapter 2) against real donation-solicitation language at a scale the field has not previously had for this domain. The project’s explicit reporting discipline – ceilings, confidence intervals, significance tests, and a stated causal limitation rather than a suppressed one – is offered as a template for how this kind of applied, corpus-driven persuasion research should be reported, independent of the specific numbers obtained here.

Chapter 7

Conclusion

7.1 Summary of Contributions

This project set out to answer two questions about persuasive donation-solicitation dialogue that the public Persuasion-For-Good corpus could not previously answer at scale: which persuasion strategies a persuader actually uses, turn by turn, and which of those strategies are associated with a successful outcome. In addressing them it produced:

1. A 41-strategy / 11-category multi-label persuasion taxonomy, and an exhaustive, disclosed, LLM-assisted-and-human-checkpointed annotation of all 10,600 persuader turns (1,017 dialogues) of the corpus against it (Section 4.2), together with a direct measurement of that annotation process’s own reliability (Section 5.2) so that every downstream result can be judged against an honest ceiling.
2. A multi-label strategy classifier reaching category-level **micro-F1 0.706** / **macro-F1 0.564** on held-out test data – within 9 points of the corpus’s own measured cross-annotator agreement ceiling (0.793) – reached through seventeen documented, hypothesis-driven, sentinel-verified design iterations (Chapter 5), two of which contribute reusable, empirically-validated ideas beyond this specific project: an empirical-Bayes threshold-shrinkage decision rule that out-performs the standard per-label-optimal rule under class imbalance (Section 5.5), and a demonstrated, measured case where task reframing (making a coarser, better-supported label set the primary training objective rather than a derived by-product) outperforms further tuning of the finer-grained task directly (Section 4.5).
3. A donation-outcome analysis on gold strategy labels quantifying, with cross-validated goodness-of-fit and multiple-testing-corrected significance, which strategies are associated with donation (reciprocity-based appeals: odds ratio 4.83; threat-and-pressure tactics: odds ratio 0.40) and a controlled comparison isolating a genuine, if modest, $+0.03$ pseudo- R^2 advantage for multi-label over single-label strategy representations of the same data (Section 5.6).
4. A companion donation-intent classifier (89.5% accuracy, 0.862 macro-F1) built on a dual-stream cross-attention architecture that was further shown to transfer,

without domain-specific modification, to a structurally distinct buyer–seller negotiation domain at 97.3% accuracy (Sections 5.8–5.9).

7.2 Limitations

Four limitations are stated explicitly, because a result reported without its limitations is a result overstated.

The rare-category gap is unresolved, not solved. Three categories making up 3.4% of positive labels score a mean F1 of 0.22 against 0.69 for the rest, and this gap accounts for essentially the entire distance between the final result and the annotation ceiling (Table 5.5). The strongest lever found (equal-weight multi-task supervision) helps this specific gap; the strongest *headline*-improving lever found (the final pool expansion) does not, and in fact costs a little of it back.

The annotation is a single process, once. Reliability is measured via near-duplicate agreement within the produced annotation, not via a fully independent second annotation of the same turns by a different process; the reported ceiling is therefore a lower bound on true reliability variance, not a complete picture of it.

The outcome analysis is observational. Table 5.10’s associations are not causal effects, and the project states directly (Section 5.6) that at least one headline-looking association in earlier working notes was found, on a preliminary check, to shrink substantially once temporal ordering within the dialogue was accounted for. This report chose not to include an under-powered causal analysis rather than present a fragile result as settled.

Single-domain scope. The strategy taxonomy and its annotated corpus cover one specific solicitation context (a fixed charity, a fixed crowd-worker task design). Module D’s cross-domain result shows the *classification architecture* generalizes; it does not by itself demonstrate that the specific strategy taxonomy or the specific strategy–outcome associations in Table 5.10 would hold in a materially different solicitation context.

7.3 Future Work

Temporal causal-direction audit. Locate each dialogue’s commitment turn and re-fit the outcome models (Section 5.6) on strategies observed strictly *before* that point, and fit a discrete-time hazard model treating each turn as an at-risk observation. This directly resolves the limitation named in Section 5.6 and was the highest-priority item identified in the team’s own working notes on Module C.

Closing the rare-category gap. Targeted approaches beyond the loss-weighting and pool-composition levers already tried in this report (Section 5.7) – e.g. few-

shot or retrieval-augmented scoring specifically for the three weakest categories, or additional targeted annotation of dialogues likely to contain them.

Measurement-error-aware outcome modeling. Re-fit Module C’s models using the classifier’s *predicted* strategy labels with an explicit attenuation correction for prediction error, now that Module B’s reliability is precisely measured (Section 5.2) – uniquely possible because this project owns both the classifier and the outcome model.

Strategy-repertoire dynamics. Model adaptivity and escalation within a dialogue (does a persuader change strategy after an early refusal?) as dialogue-level predictors of outcome, a question that is only answerable once strategies are recovered at the multi-label, turn level this project provides.

Broader domain and language coverage. Extend the taxonomy-driven annotation methodology to further solicitation domains and languages, and extend Module D’s synthetic-corpus methodology to a larger, more diverse set of persuasive-dialogue domains to more thoroughly test the dual-stream architecture’s generality.

7.4 Closing Remark

The central engineering lesson of this project generalizes past its specific numbers: under a hard, externally-imposed compute budget, the value of a single well-motivated, properly-controlled experiment – with a stated hypothesis, a reproducibility check, and an honest verdict – consistently exceeded the value of broad, undirected tuning. Several of this report’s most confident-sounding, plausible design ideas measurably failed (Table 5.11); the fact that this is known, rather than silently absent from the record, is the result this project is most confident stands on its own.

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